Lesson notes · for you

Whole numbers only

Optimization · Level 2 · Lesson 3

Time
20 min

Materials
Six paper cards and a pen; A box or a square drawn on paper for the truck

Learning goals

- Explain why some choices come only whole, or yes or no.
- Compare a quick (greedy) rule with the best load.
- Show why rounding can break a limit.

Activity: Beat the quick rule

1. Cards (4 min): Copy the six truck loads onto cards (solar 6 and 42, desks 5 and 33, books 5 and 32, filter 3 and 15, footballs 2 and 8, chalk 1 and 3, weights in hundreds of kilos), or make your own.
2. Greedy (4 min): Work out points per 100 kg for each card, sort them, and pack from the top while they fit under 10. Write the total.
3. Search (8 min): Try other loads that fit and write each total. Look for one that beats the quick rule.
4. Compare (4 min): Compare with a partner. Did greedy miss the best? Would the village school set the same points?

Weights are in hundreds of kilos so the sums stay small: the truck holds 10.

Discussion

- Why did the solar panels make the quick rule miss?
- Where else does rounding cause trouble?
- Who should set the points when you help another school?

Check yourself

- One choice that must be whole, and what rounding would get wrong.

Try it this week
Make six cards, each with a weight and points, and choose a weight limit. Pack them with the quick rule (most points per kilo first), then try to beat it. List every load that fits if you can: there are only 64.

Worksheet

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Name _____ Class _____ Date _____

1 **64 people are going on a class trip and a bus seats 30, so the maths says 2.13 buses. Your class books 2 buses. What goes wrong?**

◯ Four people have no seat ◯ Nothing: it's the cheapest ◯ We pay for 0.13 of a bus

2 **The desks (500 kg) and the books (500 kg) can go on the 1,000 kg truck together.**

◯ True ◯ False

3 **Which load gives the most points per 100 kg, so the quick rule takes it first?**

◯ Chalk and paper: 3 points for 100 kg ◯ Solar panels: 42 points for 600 kg

◯ Desks: 33 points for 500 kg ◯ Water filter: 15 points for 300 kg

4 **Match each choice to the kind of number it needs.**

How many buses for the trip	• • • •	Any amount, fractions too
Whether the footballs go on the truck	• • • •	A whole number: 0, 1, 2, …
Litres of water for the garden	• • • •	Yes or no: 0 or 1

5 **Which of these loads fit on the 1,000 kg truck? Choose all that apply.**

☐ Solar panels and desks ☐ Desks and books ☐ Solar panels, filter and chalk

☐ Solar panels and books ☐ Books, filter and footballs

✎ **Think of a choice in your life that can only be whole, or yes or no. When have you seen a quick rule like "biggest first" give a worse result?**

Check your answers

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1 **Four people have no seat**

2 × 30 = 60 seats for 64 people, so 4 stay behind: the plan breaks the seating limit.

2 **True**

500 + 500 = 1,000 kg, exactly the limit, which is allowed (at most 1,000).

3 **Solar panels: 42 points for 600 kg**

Divide points by hundreds of kilos: solar 42 ÷ 6 = 7, desks 33 ÷ 5 = 6.6, filter 15 ÷ 3 = 5, chalk 3 ÷ 1 = 3.

4 **How many buses for the trip — A whole number: 0, 1, 2, …**
Whether the footballs go on the truck — Yes or no: 0 or 1
Litres of water for the garden — Any amount, fractions too

Buses only come whole, a load is taken or not, and water can be measured in any amount, so a fraction like 2.5 litres is fine.

5 **Desks and books; Solar panels, filter and chalk; Books, filter and footballs**

Add the weights: desks and books 1,000; solar, filter and chalk 600 + 300 + 100 = 1,000; books, filter and footballs 500 + 300 + 200 = 1,000.